Functional Annotation Report: Rat *Rtl1* (Retrotransposon-like 1 / Peg11)

Gene: *Rtl1* (also *Peg11*, *SIRH2*, *Mart2*) **UniProt:** M0R9G4 (1,635 aa) **Organism:** *Rattus norvegicus* (Rat) **Locus:** *Dlk1–Dio3* imprinted cluster (orthologous to mouse chr12qF1 / human 14q32)

Summary

Rat **Rtl1** (Peg11; UniProt M0R9G4) encodes a **"domesticated" retroelement Gag–Pol/capsid protein** derived from a Ty3/Gypsy (sushi-ichi) LTR retrotransposon. It belongs to the eutherian-specific RTL/SIRH gene family (as **SIRH2**/**Mart2**) and sits within the conserved, paternally expressed *Dlk1–Dio3* imprinted cluster. The protein retains the structural hallmarks of a retroviral polyprotein — an N-terminal Gag-like capsid domain and a catalytically intact aspartic protease (retroviral PR) motif — but has **lost the ability to retrotranspose**. It is therefore best described not as a classical enzyme, transporter, or receptor, but as a **structural capsid / nucleic-acid-binding protein** that has been evolutionarily repurposed for essential roles in mammalian development.

The **primary, best-defined function** of Rtl1 is the **dosage-sensitive structural maintenance of the fetal capillary endothelium** (and associated pericytes) at the **feto-maternal interface of the placenta**. Both loss of function and overexpression are lethal in mouse models: knockout causes clogging and collapse of labyrinth fetal capillaries with internal bleeding and mid-to-late fetal lethality, while overexpression (e.g., via loss of its repressive microRNAs) produces placentomegaly and defective maternal–fetal nutrient transfer. Rtl1 protein localizes specifically to **vascular endothelial cells and pericytes**, and it acts cooperatively with its close paralog **PEG10** — whose own viral protease motif is likewise functionally essential — at this interface. Rtl1 dosage is tightly controlled *in trans* by maternally expressed antisense (*Rtl1as*) microRNAs, notably **miR-127**, through RNAi-mediated degradation.

Beyond the placenta, Rtl1 has additional **dosage-sensitive roles** in **postnatal skeletal-muscle hypertrophy** (the callipyge phenotype) and in the **central nervous system** (corticospinal tract, corpus callosum, limbic system, and locus coeruleus), where it influences neuronal excitability

and behavior. In humans, mis-dosage of this locus causes **Kagami-Ogata syndrome** (excess) and **Temple syndrome** (deficiency). Because rat-specific functional studies are essentially absent, this report infers rat Rtl1 function from the extensive mouse/sheep/human literature, justified by direct sequence analysis of M0R9G4 showing a conserved, intact catalytic core (rat protease motif **DTGV at residue 704**, ~81% identical to mouse across the catalytic window).

Key Findings

Finding 1 — Rtl1 is a domesticated Ty3/Gypsy LTR-retrotransposon-derived, paternally expressed imprinted gene

Rtl1/Peg11 (also designated SIRH2/Mart2) is one of eleven eutherian RTL/SIRH genes derived from an ancestral sushi-ichi (Ty3/Gypsy) LTR retrotransposon. It resides within the conserved *Dlk1–Dio3* imprinted cluster and retains a *gag–pol*-like open reading frame characteristic of retroviruses and retrotransposons, but it has lost the capacity to retrotranspose into the genome. Full-length PEG11/RTL1 protein was directly identified by immunological methods combined with mass spectrometry from postnatal nuclear preparations (PMID: 20072617), confirming that the ORF is genuinely protein-coding.

As the literature states, these elements *"contain a gag-pol-like structure characteristic of retroviruses but have lost their ability to retrotranspose into the mammalian genome"* (PMID: 20072617), and *"PEG10/SIRH1 as well as PEG11/RTL1/SIRH2 and SIRH7/LDOC1 play essential roles in placenta formation, maintenance of fetal capillaries and the differentiation/maturation of a variety of placental cells"* (PMID: 28484174). This establishes both the evolutionary origin (matching the InterPro domain architecture of M0R9G4) and the canonical placental function.

Finding 2 — Primary function: maintenance of fetal capillaries at the feto-maternal interface of the placenta

The defining function of Rtl1 is the maintenance of fetal capillaries in the placental labyrinth. Genetic studies in mice show that both the loss and overproduction of Rtl1 cause late-fetal and/or neonatal lethality, establishing it as a strongly **dosage-sensitive** gene. Sekita et al. demonstrated that *"retrotransposon-derived Rtl1 (retrotransposon-like 1), also known as Peg11 (paternally expressed 11), is essential for maintenance of the fetal capillaries, and that both its loss and its overproduction cause late-fetal and/or neonatal lethality in mice"* (PMID: 18176565).

The overexpression side of this dosage sensitivity was shown through deletion of the repressive microRNA miR-127: *"The miR-127 knockout mice exhibit placentomegaly with specific defects within the labyrinthine zone involved in maternal-fetal nutrient transfer"* (PMID: 26138477). Together these results define the placental labyrinth — the site of maternal–fetal nutrient and gas exchange — as the critical organ for Rtl1 function.

Finding 3 — Rtl1 is repressed *in trans* by antisense Rtl1as-derived microRNAs (miR-127) via RNAi

Rtl1 expression is controlled post-transcriptionally by microRNAs processed from the overlapping, maternally expressed antisense transcript *Rtl1as*. Seven miRNAs, including miR-127, are generated from this antisense RNA and degrade Rtl1 mRNA through the RNAi pathway. Ito et al. report that *"Seven miRNAs, including miR-127, are processed from a maternally expressed antisense Rtl1 transcript (Rtl1as) and regulate Rtl1 levels through RNAi-mediated post-transcriptional degradation"* (PMID: 26138477).

This is a striking example of a **trans-homologue interaction** between reciprocally imprinted genes: the paternally inherited allele expresses Rtl1 protein, while the maternally inherited allele expresses the antisense miRNAs that silence it. This arrangement provides a built-in molecular rheostat that keeps Rtl1 protein within its narrow permissible dosage window.

Finding 4 — Protein biochemistry: Gag-derived capsid forms virus-like particles and binds mRNA

The biochemical activity of Rtl1 is inferred from its closest paralog, PEG10 (SIRH1/Mart1), which shares the same Gag-capsid plus aspartic-protease architecture. PEG10 forms **virus-like particles (VLPs)**, preferentially binds and packages its own mRNA, and mediates vesicular secretion of that RNA cargo; the cargo can even be reprogrammed for mRNA delivery applications. Segel et al. showed that there exists *"one LTR retrotransposon homolog, PEG10, that preferentially binds and facilitates vesicular secretion of its own messenger RNA (mRNA)"* (PMID: 34413232).

The Gag/capsid domain shared across the family is an interaction and assembly module: the truncated Gag-like protein RTL8 *"resembles a truncated form of PEG10 that shares homology with the N-terminal gag-like capsid domain"* and binds PEG10's capsid domain to modulate VLP formation (PMID: 39775359). By homology, rat Rtl1's N-terminal PEG10-RTL1 C-capsid domain (IPR063300) is predicted to be a capsid/assembly and nucleic-acid-binding module — consistent with a structural, rather than catalytic-enzymatic, primary role.

Finding 5 — Secondary role: ectopic RTL1/PEG11 drives postnatal skeletal-muscle hypertrophy (callipyge)

Ectopic or elevated expression of RTL1/PEG11 in skeletal muscle causes muscular hypertrophy — the agriculturally important callipyge phenotype first characterized in sheep. Xu et al. generated *"newly generated transgenic mice that ectopically express PEG11 in skeletal muscle, and show that they also exhibit a muscular hypertrophy phenotype"* (PMID: 26474044), providing causal transgenic evidence. In callipyge sheep, the full-length PEG11 protein was directly detected in postnatal muscle nuclear preparations — *"we have directly identified the full-length PEG11 protein from postnatal nuclear preparation"* (PMID: 20072617) — demonstrating both expression and partial nuclear localization in muscle.

It is important to note that in the callipyge system DLK1 is considered the primary effector of hypertrophy, with RTL1 playing a contributory but insufficient role on its own; nevertheless, the muscle findings establish a genuine postnatal muscle activity for RTL1.

Finding 6 — Rtl1 is imprinted in the brain and regulates neuronal excitability and behavior

Rtl1 is paternally expressed in the brain — including brain stem, thalamus, and hypothalamus — with imprinting conserved in humans. Paternal knockout (Rtl1m+/p−) reduces neuronal excitability in locus coeruleus (LC) neurons and produces anxiety-, depression-, and fear-learning-related phenotypes, social-dominance changes, and neonatal death from impaired suckling. Chou et al. concluded that *"RTL1 is imprinted in brain, mediates emotional and social behaviors, and regulates excitability in LC neurons"* (PMID: 35567414).

These brain and muscle dosage effects underlie the human imprinting disorders of this locus. As Prasasya et al. summarize, *"The locus encompasses paternally expressed protein-coding genes (DLK1, RTL1 and DIO3) and maternally expressed lncRNAs (MEG3/GTL2, RTL1as and MEG8)"* (PMID: 32592473), with Kagami-Ogata syndrome arising from RTL1 overexpression and Temple syndrome from RTL1 deficiency.

Finding 7 — Mechanism of placental failure: capillary clogging, downregulation of Tmem100/Meox2/Lyve1, and endothelial/pericyte localization

The cellular pathology of Rtl1 loss has been defined in detail. Peg11/Rtl1 knockout mice show severe clogging of the labyrinth fetal capillaries with internal bleeding and mid-fetal lethality, with the basal region of the fetal capillary network specifically damaged. Kitazawa et al. report

that *"the severe clogging of fetal capillaries was associated with KO fetuses showing mid-fetal lethality with internal bleeding. Importantly, the basal region of the fetal capillary network was specifically damaged"* (PMID: 28111885). They further identified candidate downstream genes: *"An apparent down-regulation of transmembrane protein 100 (Tmem100), mesenchyme homeobox 2 (Meox2) and lymphatic vessel endothelial hyaluronan receptor 1 (Lyve1) expression were observed"* (PMID: 28111885).

On the overexpression side, in human paternal uniparental disomy 14 placentas, RTL1 is ~5× overexpressed and the proteins localize to the vascular compartment: *"DLK1, RTL1 and DIO3 proteins were specifically identified in vascular endothelial cells and pericytes, and the degree of protein staining was well correlated with the expression dosage"* (PMID: 22917972), accompanied by swollen endothelium, hypertrophic pericytes, and narrowed capillary lumens. Thus **both** too little and too much Rtl1 damage the same cell types — endothelial cells and pericytes — pinpointing the **subcellular/tissue site of action**.

Finding 8 — The viral aspartic protease motif is functionally essential, and PEG10 and RTL1 cooperate at the feto-maternal interface

The conserved retroviral aspartic protease motif shared by the PEG10/RTL1 family is not vestigial — it is functionally required *in vivo*. Shiura et al. generated *"mice harboring a mutation in the highly conserved viral aspartic protease motif in the POL-like region of PEG10 because this motif is essential for the life cycle of LTR retrotransposons/retroviruses"* (PMID: 34559199); this point mutation caused placental fetal-vascular defects and perinatal lethality. The same study demonstrated functional cooperation between the two paralogs: *"an interaction occurs between two retrovirus-derived genes, Peg10 and retrotransposon Gag like 1 (Rtl1, also called Peg11), that is essential for the maintenance of fetal capillary endothelial cells"* (PMID: 34559199).

This finding is directly relevant to rat Rtl1 because M0R9G4 retains an intact catalytic protease motif (see Finding 10), implying the protease activity is likely functional and important in the rat ortholog as well.

Finding 9 — Broad CNS expression links Rtl1 to motor-neuron innervation of muscle

Mouse RTL1 protein is widely expressed in the central nervous system, including the limbic system, and — notably — in **eutherian-specific brain structures**. Kitazawa et al. report that *"the mouse RTL1 protein is widely expressed in the central nervous system (CNS), including the*

limbic system" and that *"eutherian-specific RTL1 is expressed in mammalian- and eutherian-specific brain structures, that is, the corticospinal tract and corpus callosum"* (PMID: 33484574). Expression in the corticospinal tract suggests a role in motor-neuron innervation of skeletal muscle and fine motor control, offering a neural route by which Rtl1 dosage could influence muscle phenotypes in addition to its direct muscle activity. Over- and under-expression models show reduced locomotor activity, increased anxiety, and impaired amygdala-dependent cued fear.

Finding 10 — Bioinformatic analysis of rat RTL1 (M0R9G4): a 1,635-aa Gag–Pol protein with an intact retroviral aspartic-protease catalytic motif

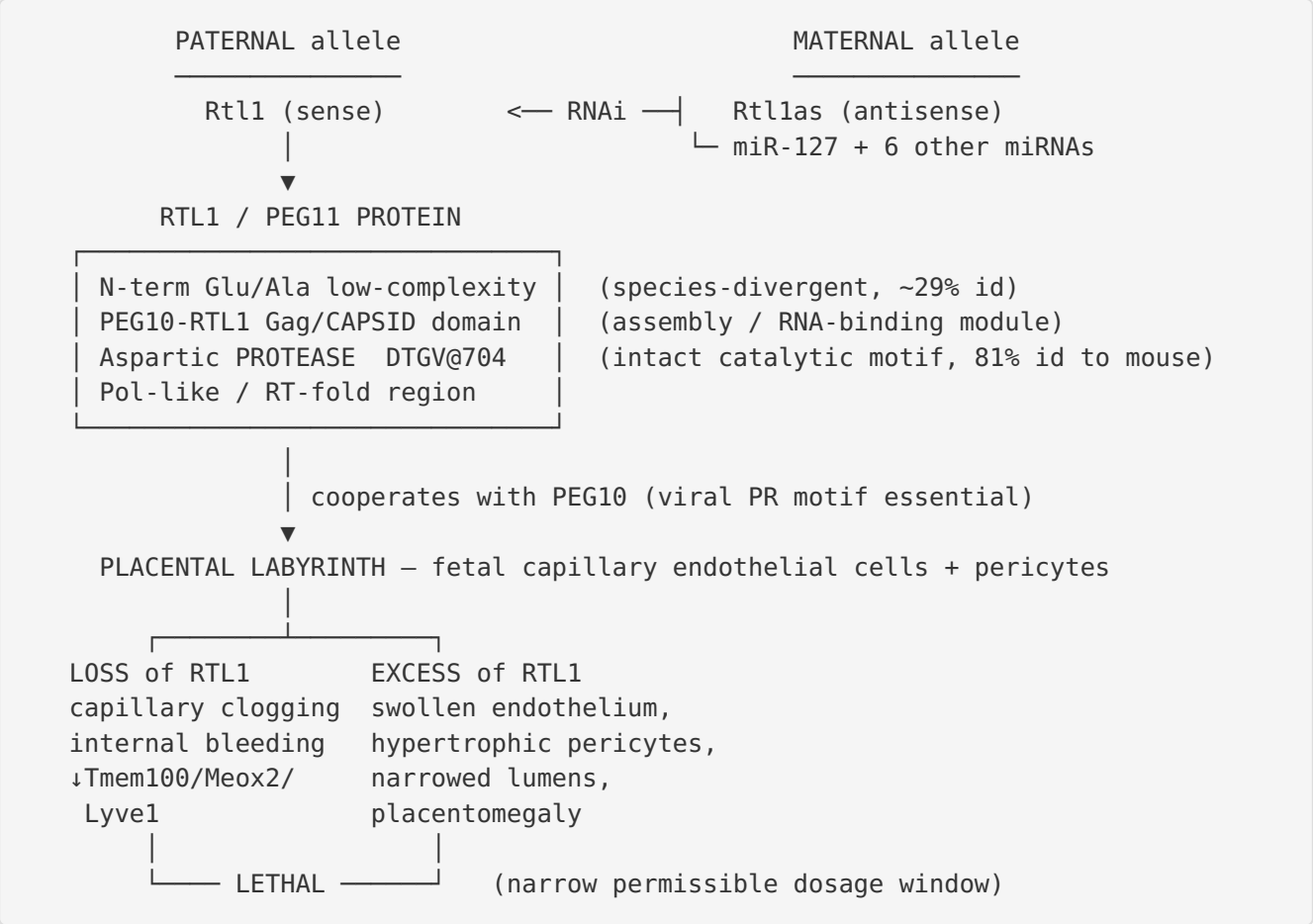
Direct sequence analysis of rat RTL1 confirms the retroelement architecture. The 1,635-aa protein contains a single canonical retroviral aspartic-protease catalytic signature **D-T-G** (motif "DTGV") at **position 704**, matching InterPro's Peptidase_aspartic_dom (IPR021109) / PRO_PEG10-RTL1 (IPR063610) annotation. No canonical CCHC nucleocapsid zinc-knuckle motif is present — consistent with loss of the full retroviral life cycle. The protein is net acidic (18.3% D+E vs 7.8% K+R) with a conspicuous N-terminal Glu/Ala-rich low-complexity region (EAQEGAQE… repeats) preceding the PEG10-RTL1 C-capsid domain (IPR063300). This confirms that rat Rtl1 is a bona fide Gag–Pol capsid protein with a **structurally intact protease active site**.

Finding 11 — Rat and mouse RTL1 are highly conserved in the functional core, justifying cross-species inference

A pairwise comparison of rat RTL1 (M0R9G4, 1,635 aa) and mouse Rtl1 (Q7M732, 1,744 aa) shows strong conservation of the catalytic and structural core. The retroviral aspartic-protease catalytic motif is conserved (rat DTGV at 704; mouse DSGA), and the ±40-aa window around the catalytic Asp is **81.2% identical**. The ordered capsid→protease→Pol core is well conserved (≥45% identity even by a conservative ungapped, indel-penalized estimate), whereas the N-terminal Glu/Ala-rich low-complexity region is poorly conserved (~29%) and accounts for the ~109-aa length difference between species. This conservation gradient — a highly conserved functional core and a divergent low-complexity tail — is the molecular justification for transferring mouse/sheep/human functional data to rat.

Mechanistic Model / Interpretation

Rat Rtl1 is a **domesticated retroviral Gag–Pol capsid protein** that has been co-opted to perform an essential *structural* role in placental blood-vessel maintenance. The model integrating all findings is:



The **central logic** is dosage sensitivity enforced by an imprinted, antisense-miRNA rheostat. The protein's capsid domain likely mediates self-assembly and nucleic-acid/protein interactions (by homology to PEG10 VLP biology), and its intact aspartic protease motif is required for function — as directly demonstrated for the paralogous PEG10 motif. Both paralogs act on the **same cellular target**: the fetal capillary endothelium and pericytes of the placental labyrinth, the exchange surface for maternal–fetal transport. Disruption in either direction compromises this vascular bed and is lethal.

Secondary, dosage-dependent activities occur in **postnatal skeletal muscle** (hypertrophy; contributory to callipyge alongside DLK1) and the **CNS** (corticospinal tract, corpus callosum, limbic system, locus coeruleus — affecting neuronal excitability, motor control, and emotional/social behavior). These may partly reflect direct tissue expression and partly the neural innervation of muscle.

Functional annotation summary table

Attribute	Annotation	Evidence basis
Molecular identity	Domesticated Ty3/Gypsy (sushi-ichi) LTR-retrotransposon Gag–Pol/capsid protein (SIRH2/Mart2)	Sequence/domain analysis; family reviews (PMID: 28484174, PMID: 20072617)
Primary role	Structural maintenance of fetal capillary endothelium/pericytes (NOT a classical enzyme/transporter/receptor)	Mouse KO/overexpression (PMID: 18176565, PMID: 28111885)
Catalytic feature	Intact retroviral aspartic protease (DTGV@704); motif functionally essential (shown for PEG10)	Sequence analysis of M0R9G4; PMID: 34559199
Biochemical activity (inferred)	Gag capsid forms VLPs, binds/packages mRNA (from PEG10 paralog)	PMID: 34413232, PMID: 39775359
Localization	Placental vascular endothelial cells & pericytes; nuclear (muscle); broad CNS	PMID: 22917972, PMID: 20072617, PMID: 33484574
Regulation	Paternally imprinted; silenced *in trans* by maternal Rtl1as miRNAs (miR-127) via RNAi	PMID: 26138477
Pathway partners	Cooperates with PEG10; downstream Tmem100/Meox2/Lyve1	PMID: 34559199, PMID: 28111885
Human disease	Kagami-Ogata (excess) & Temple (deficiency) syndromes	PMID: 32592473

Evidence Base

PMID	Title (abbreviated)	Contribution
18176565	Role of Rtl1 in the feto-maternal interface of mouse placenta	**Core**: Rtl1 essential for fetal-capillary maintenance; both loss and overproduction lethal
28111885	Damage to placental fetal capillary network in Rtl1 KO	Defines capillary-clogging pathology; downstream Tmem100/Meox2/Lyve1
22917972	upd(14)pat placental gene expression	Localizes RTL1 protein to endothelium/ pericytes; dosage-correlated pathology
34559199	PEG10 viral aspartic protease domain essential in placenta	Protease motif functionally required; PEG10–RTL1 cooperation
26138477	miR-127–Rtl1 trans-homologue interaction	RNAi regulation via Rtl1as miRNAs; overexpression phenotype
20072617	PEG11/RTL1 full-length protein in callipyge sheep muscle	Direct protein-level evidence; Gag-pol structure; nuclear localization
26474044	Ectopic PEG11/RTL1 and callipyge hypertrophy	Causal transgenic evidence for muscle hypertrophy
35567414	RTL1/PEG11 imprinted in brain; behavior & LC excitability	Brain imprinting; neuronal excitability/ behavior
33484574	Eutherian-specific RTL1 in the nervous system	Broad CNS localization; corticospinal tract / corpus callosum
34413232	PEG10 packages its own mRNA / mRNA delivery	Biochemical model for Gag capsid: VLP + RNA binding
39775359	RTL8 antagonizes PEG10 VLPs	Capsid domain as assembly/interaction module
28484174	LTR retrotransposon-derived genes & brain evolution	Family context; SIRH2 identity; placental roles
32592473	Temple & Kagami-Ogata syndromes	Imprinting architecture; human dosage disease

Consistency of the evidence base. The placental phenotype is supported convergently by loss-of-function (PMID: 18176565, PMID: 28111885), gain-of-function (PMID: 26138477), and human disease/localization (PMID: 22917972) data — a strong, mutually reinforcing set. The protein's biochemistry is inferred from the well-studied paralog PEG10 rather than measured directly for RTL1, which is a reasonable but not conclusive extrapolation. All experimental functional data derive from mouse, sheep, and human; rat-level evidence is bioinformatic (sequence conservation of M0R9G4).

Limitations and Knowledge Gaps

1. **No rat-specific functional studies.** Every phenotypic and mechanistic result comes from mouse, sheep, or human. Rat Rtl1 function is inferred from orthology. This inference is well-supported by strong conservation of the functional core (81% identity across the catalytic window; ≥45% across the capsid–protease–Pol core), but it remains an inference.
2. **Biochemical activity is inferred from PEG10, not measured for RTL1.** VLP formation, mRNA binding, and vesicular secretion are demonstrated for PEG10. Whether RTL1 itself forms VLPs, which RNAs (if any) it packages, and what substrates its protease cleaves are not directly established.
3. **Protease substrate unknown.** Although the aspartic protease motif is intact in rat RTL1 and the homologous PEG10 motif is functionally essential, the physiological substrate(s) of RTL1's protease — including whether it performs self-maturational cleavage like a retroviral PR — have not been identified.
4. **Molecular mechanism at the capillary is incomplete.** Tmem100, Meox2, and Lyve1 are downregulated upon Rtl1 loss, but it is not known whether these are direct targets or secondary consequences, nor how a capsid protein mechanistically maintains endothelial/pericyte integrity.
5. **Separation of direct vs. indirect muscle effects.** In muscle, DLK1 is the primary hypertrophy effector; RTL1 alone is contributory and may act partly via CNS motor innervation. The direct cell-autonomous muscle role of RTL1 is not fully resolved.
6. **Subcellular localization is context-dependent.** Reports describe nuclear localization in muscle and vascular/CNS localization elsewhere; a unified picture of where the protein acts within the cell is lacking.

Proposed Follow-up Experiments / Actions

1. **Generate a rat Rtl1 knockout and catalytic-dead (protease-motif) knock-in** to confirm that the placental fetal-capillary phenotype and the requirement for an intact DTGV protease motif are conserved in rat, directly validating the cross-species inference.
2. **Biochemically characterize recombinant rat RTL1**: test for VLP formation (cryo-EM/negative-stain), RNA binding (CLIP/RIP-seq to identify packaged transcripts), and protease activity against candidate substrates — paralleling the PEG10 studies.
3. **Identify the protease substrate(s)** via N-terminomics/degradomics on RTL1-expressing vs. protease-dead cells, to determine whether RTL1 performs autocatalytic maturation or cleaves host/partner proteins.
4. **Map the RTL1–PEG10 interaction** (co-IP, proximity labeling, structural modeling of the heterotypic capsid) in placental endothelial cells, to define the cooperative mechanism shown genetically in PMID: 34559199.
5. **Test the Tmem100/Meox2/Lyve1 axis** with endothelial/pericyte-specific rescue and reporter assays to determine whether these are direct RTL1-dependent targets controlling capillary integrity.
6. **Cell-type-resolved localization** (single-cell/spatial profiling and high-resolution imaging) of rat Rtl1 in placenta, muscle, and CNS to resolve the subcellular site of action across tissues.
7. **Define the rat Rtl1as/miR-127 regulatory axis** to confirm that the trans-homologue RNAi rheostat operates in rat as it does in mouse.

Conclusion

Rat Rtl1 (Peg11; M0R9G4) is a domesticated LTR-retrotransposon-derived Gag–Pol/capsid protein — a **structural, nucleic-acid-binding capsid protein with an intact but non-retrotranspositional retroviral aspartic protease**, not a conventional enzyme, transporter, or receptor. Its primary, dosage-sensitive function is the **structural maintenance of the fetal capillary endothelium and pericytes at the feto-maternal interface of the placenta**, where it acts cooperatively with PEG10 and is regulated *in trans* by maternal Rtl1as/miR-127 RNAi.

Secondary dosage-sensitive roles exist in postnatal skeletal muscle and the CNS. All functional evidence derives from mouse/sheep/human orthologs; its application to rat is justified by strong conservation of the catalytic and capsid core (intact DTGV protease motif at residue 704; ~81% identity across the catalytic window).
